Java 1D Array Coding Questions (Basic)

1. Input & Print Array

- Write a Java program to take **N elements as input** into an array and print them.

2. Find Maximum and Minimum Element

- Write a program to find the **largest and smallest element** in an array.

3. Sum & Average of Array Elements

- Find the **sum** and **average** of all elements in an array.

4. Reverse an Array

- Write a program to **reverse** a given array (without using extra array).

5. Linear Search

- Implement a program to **search** for an element in an array (linear search).

6. Count Even and Odd Numbers

- Count how many **even** and **odd** numbers are present in an array.

7. Copy Array

- Copy all elements from one array into another array.

8. Second Largest Element

- Find the **second largest element** in an array.

9. Frequency of Elements

- Write a program to count the **frequency of each element** in an array.

10. Check if Array is Sorted

- Check whether the given array is **sorted in ascending order**.

11. Merge Two Arrays

- Merge two arrays into a **single array**.

12. Remove Duplicates

- Remove **duplicate elements** from a sorted array.

13. Rotate Array

- Rotate an array to the **left** or **right** by k positions.

14. Find Missing Number (1 to N)

- Given an array containing numbers from 1 to N with one missing, find the **missing number**.

15. Find Duplicate Element

- Find the first **duplicate element** in an array.

Java 2D Array Coding Questions (Basic to Medium)

1. Input & Print 2D Array

- Write a Java program to **take input for a matrix** and print it in row & column format.

2. Sum of All Elements

- Find the **sum of all elements** in a 2D array.

3. Row-wise & Column-wise Sum

- Print the **sum of each row** and **sum of each column** separately.

4. Maximum Element in Matrix

- Find the **largest element** in a 2D array.

5. Transpose of a Matrix

- Write a program to find the **transpose of a matrix**.

6. Matrix Multiplication

- Multiply two matrices and print the resulting matrix.

7. Diagonal Elements

- Print the **main diagonal** and **secondary diagonal** elements of a square matrix.
- Find their sum separately.

8. Upper & Lower Triangle of Matrix

- Print elements of the **upper triangular** and **lower triangular** part of a square matrix.

9. Identity Matrix Check

- Write a program to check if a given square matrix is an **identity matrix**.

10. Sparse Matrix Check

- Count the number of **zero elements** in a matrix and check if it is a **sparse matrix**.

11. Search Element in Matrix

- Search a given element in a **2D matrix**.

12. Rotate Matrix

- Rotate a matrix **90°, 180°, or 270°** clockwise.

13. Matrix Symmetry Check

- Check whether a matrix is **symmetric** (i.e., matrix == transpose).

14. Add Two Matrices

- Write a program to **add two matrices**.